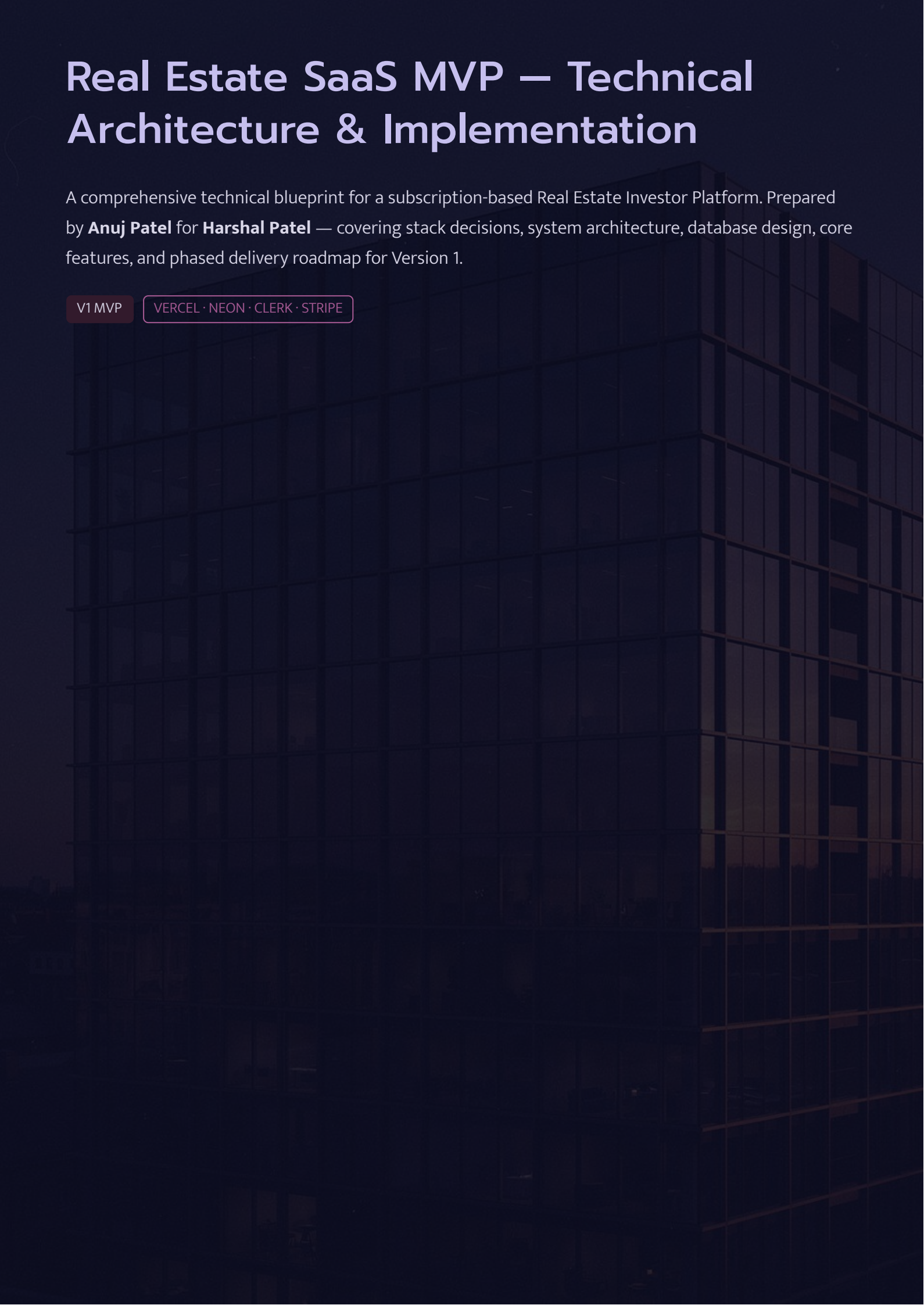
Real Estate SaaS MVP — Technical Architecture & Implementation

A comprehensive technical blueprint for a subscription-based Real Estate Investor Platform. Prepared by **Anuj Patel** for **Harshal Patel** — covering stack decisions, system architecture, database design, core features, and phased delivery roadmap for Version 1.

V1 MVP

VERCEL · NEON · CLERK · STRIPE



Project Overview

The platform is a modern, fast, and scalable Real Estate SaaS application purpose-built for property investors across multiple strategies. It enables users to analyse deals, manage pipelines, generate reports, and access AI-powered insights — all under a subscription model.



Target Users

- Wholesalers
- Fix & Flip Investors
- Buy & Hold Investors
- Rental Property Investors
- Real Estate Entrepreneurs



Core Capabilities

- Analyse investment properties
- Save & manage deals
- Track investment performance
- Generate PDF reports
- AI-powered deal insights



Platform Model

- Monthly & annual subscriptions
- Free trial support
- Coaching session bookings
- Role-based access control
- Scalable SaaS architecture

Technology Stack

The stack is chosen for type-safety, developer velocity, and seamless Vercel-native deployment. Every layer — from frontend to database — is optimised for the SaaS use case.

Frontend

- Next.js 15 (App Router)
- React + TypeScript
- Tailwind CSS + shadcn/ui
- Framer Motion
- React Hook Form + Zod

Backend

- Next.js Server Actions
- Route Handlers (API)
- TypeScript throughout

Data & ORM

- Neon PostgreSQL (serverless)
- Prisma ORM — type-safe queries
- Automatic scaling & branching

Services

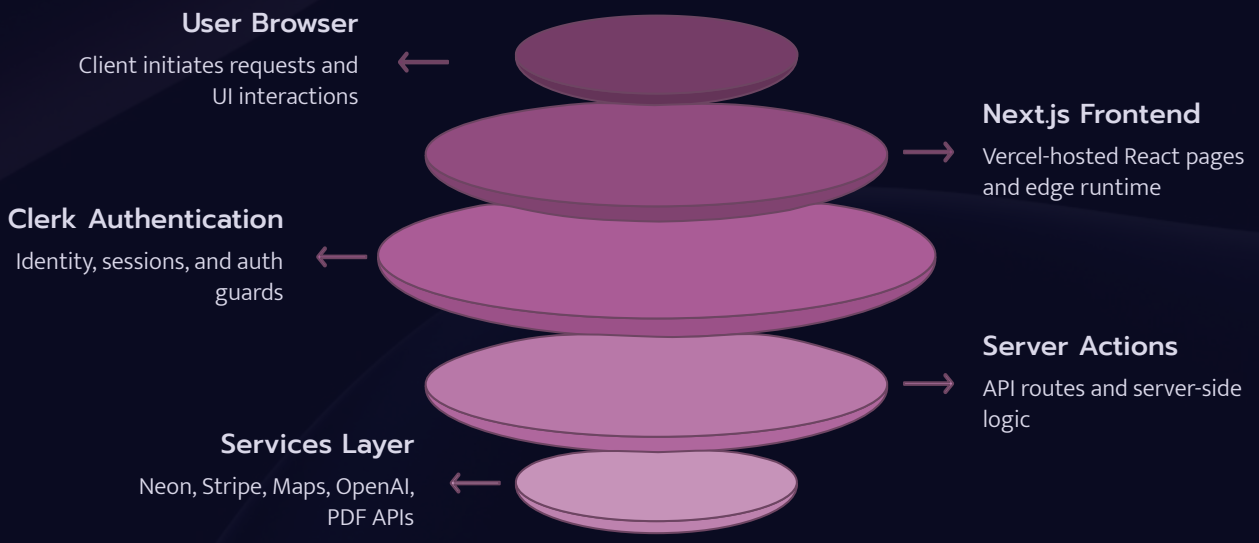
- Clerk — Authentication & sessions
- Stripe — Billing & subscriptions
- OpenAI — AI deal analysis
- Google Maps — Location context

Deployment

- Vercel (frontend & backend)
- Neon (database)

System Architecture

The architecture follows a clean, layered request flow — from the browser through Clerk authentication, into Next.js server actions, and out to specialised downstream services. All compute runs on Vercel; all persistence on Neon.



The services layer is intentionally modular — each integration (database, payments, AI, maps) is independently replaceable, supporting future V2 expansions without architectural rewrites.

User Roles & Permissions

Investor Role

- Register, log in & subscribe to a plan
- Analyse properties and run calculators
- Save deals, manage CRM pipeline
- Generate PDF reports & use AI tools
- Book coaching sessions via Calendly

Admin Role

- View, search, suspend & activate users
- Manage subscriptions & pricing plans
- Monitor AI usage & platform analytics
- Manage coaching bookings & sessions
- Configure APIs, Stripe & content CMS

Database Design

The Neon PostgreSQL schema is structured around six primary domains: identity, subscriptions, properties, deal pipeline, coaching, and audit logs. Prisma ORM enforces type-safe access throughout.

Users & Profiles

`id` UUID · `clerk_id` String · `email` · `role` Enum · `phone` · `company` · `investor_type`

Subscriptions

`stripe_customer_id` · `stripe_subscription_id` · `plan` · `status` — synced via Stripe webhooks

Properties & Analysis

Address, geo-coordinates, purchase price, rehab, ARV, rent, taxes, HOA, vacancy, cash flow, ROI, cap rate, DSCR, mortgage

Flip Analysis

Purchase, rehab, ARV, financing, holding cost, selling cost, profit, ROI, MAO — full fix-and-flip underwriting

Deal Pipeline & CRM

Lead, seller, buyer, offer, assignment fee, notes, follow-up date, stage — full wholesaler workflow

Coaching & Activity Logs

Booking, payment ref, meeting link · Login events, analyses created, reports generated, subscription changes

Core Features — Version 1

V1 ships a complete end-to-end product covering public marketing, authenticated dashboards, calculators, CRM, billing, and coaching — everything an investor needs from day one.

Landing Website

- Hero, Features, Benefits
- Pricing, FAQ, Testimonials
- CTA + Login/Register

Auth & Billing

- Clerk: signup, login, reset
- Stripe: monthly, annual, trial
- Customer portal + webhooks

Investor Dashboard

- Saved & active deals
- ROI & cash flow overview
- Recent activity feed

Calculators

- Property analysis (full)
- Fix & Flip calculator
- Rental & DSCR calculator

CRM Pipeline

- Add & stage leads
- Notes & follow-ups
- Saved deals management

Coaching

- Calendly integration
- Stripe payment for sessions
- Booking management

Property Analysis Calculator

The flagship feature — a comprehensive underwriting tool that accepts raw deal inputs and outputs all critical investment metrics investors need to make confident decisions. Results are saveable and exportable to PDF.

Inputs

- Purchase Price
- Rehab Cost
- After Repair Value (ARV)
- Monthly Rent
- Down Payment %
- Property Taxes
- Insurance
- HOA Fees
- Vacancy Rate %

Calculated Outputs

- Monthly Mortgage Payment
- Monthly Cash Flow
- Return on Investment (ROI)
- Cap Rate
- Debt Service Coverage Ratio (DSCR)
- Cash-on-Cash Return

Additional Tools

- Fix & Flip: Profit, MAO, Holding Cost
- Rental: Long-term projection, Cap Rate

Advanced Features Included in V1

Beyond core calculators and CRM, V1 ships four advanced capabilities that meaningfully differentiate the platform: real property data, location intelligence, branded PDF reports, and AI-powered deal commentary.



Property APIs

Basic property lookup via Zillow, RentCast, ATTOM, or Estated — pulling property details, rent estimates, and public records to pre-populate analysis forms.



Google Maps Integration

View property locations on an interactive map, open directions, and assess neighbourhood context. Full map search is scoped to V2.



PDF Reports

Generate branded reports containing property details, full financial analysis, ROI, cap rate, and AI summary. Built with React PDF or Puppeteer.

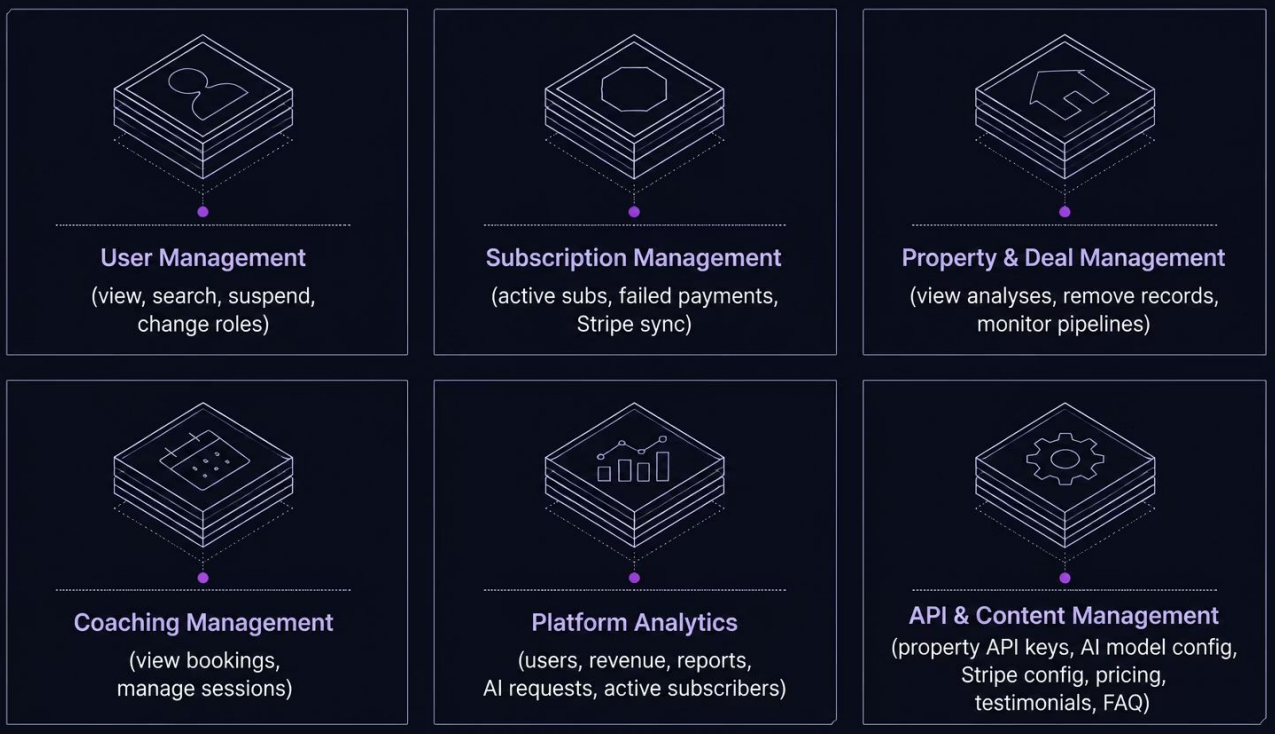


AI Deal Analysis

OpenAI-powered commentary on entered deal numbers — surfacing investment strengths, risks, and profitability insights in plain English for every analysis saved.

Admin Panel

The admin dashboard gives the platform operator full operational visibility and control — from user management and Stripe reconciliation to content updates and API configuration.



Security Architecture

Security is layered across every tier of the stack — authentication, transport, database, payments, and access control — so no single point of failure can compromise the platform or its users' data.

Identity & Access

Clerk manages all authentication flows with secure session tokens, email verification, and role-based access control (RBAC) enforced at route and server action level.

Transport & Data

All traffic served over HTTPS. Neon PostgreSQL encrypts data at rest. Stripe webhook signatures are validated server-side before processing any billing events.

API Protection

Rate limiting applied to all public API routes to prevent abuse. Authenticated routes are gated behind Clerk middleware, blocking unauthenticated requests before they reach business logic.

Performance Optimisation

The platform leverages Next.js 15's server-first model alongside database and edge strategies to deliver fast load times and a smooth user experience at scale.

Server Components

Render data-heavy views on the server, reducing client-side JS bundle size and improving Time to First Byte.

Image Optimisation

Next.js built-in `next/image` handles lazy loading, WebP conversion, and responsive sizing automatically.

Database Indexing

Prisma migrations include indexes on high-frequency query fields (user_id, status, created_at) for fast dashboard data retrieval.

Caching & Edge

Next.js fetch caching and Vercel Edge Functions minimise origin hits for static and semi-static content, reducing global latency.

Development Phases

Delivery is structured across five sequential phases, each shipping a coherent vertical slice of functionality — from initial infrastructure through to production deployment.



Project Deliverables

Upon completion, the client receives a fully deployed, production-ready platform along with all source code, infrastructure configuration, and supporting documentation for long-term ownership.

01

Platform & Source Code

Complete SaaS platform with full source code repository, Vercel deployment, Neon database, and Clerk + Stripe configuration.

02

Core Product Features

Admin dashboard, investor dashboard, property analysis calculators, CRM pipeline, saved deals, and coaching booking flow.

03

Advanced Integrations

PDF report generation, AI deal analysis (OpenAI), Google Maps integration, and basic property API lookup.

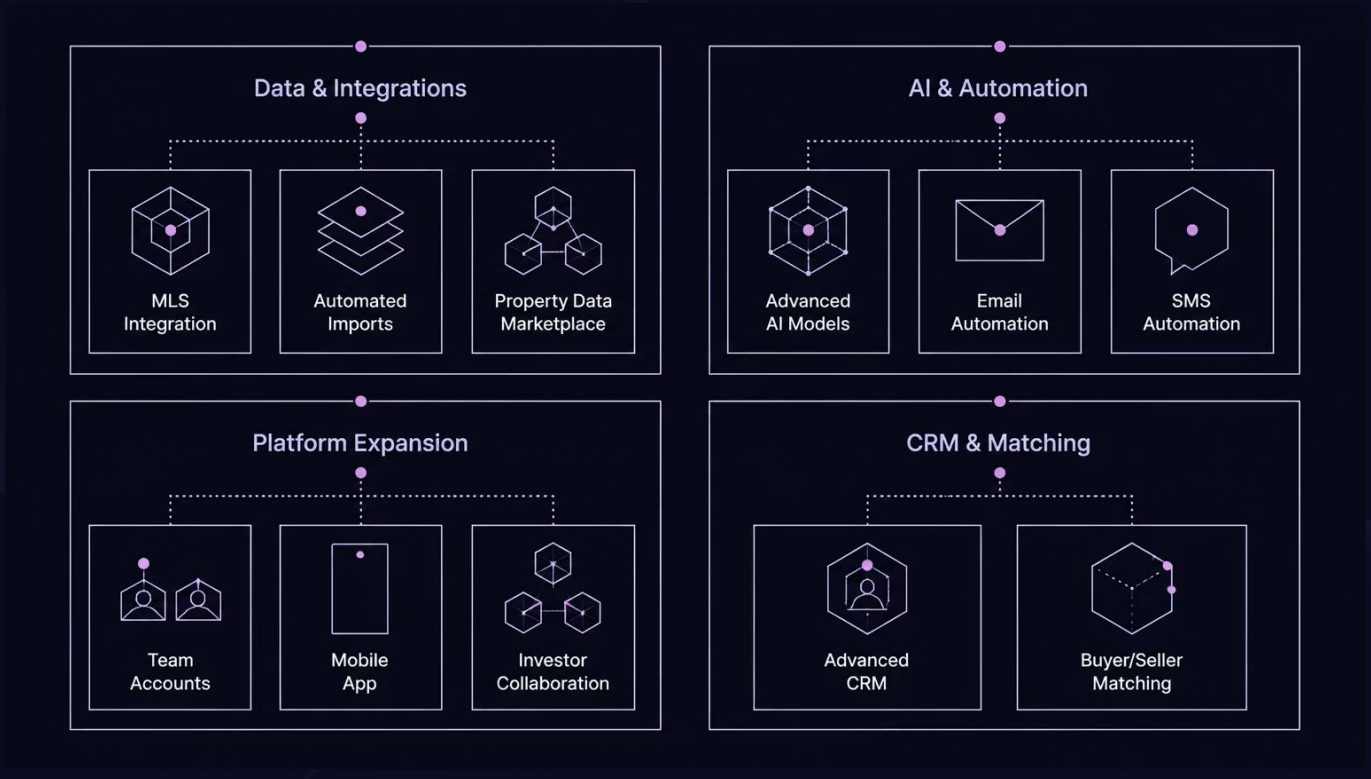
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Documentation & Handoff

Technical documentation covering architecture, environment setup, and API configuration — plus a live training session for the client team.

Future Version — V2 Roadmap

V1 is architected to support a wide range of V2 enhancements without breaking changes. The modular service layer and clean data model create a strong foundation for the features below.



V2 scope is intentionally excluded from the V1 contract. These items are captured here as architectural guardrails — all V1 decisions account for forward compatibility with these enhancements.

Strategic Rationale by Theme

Data & Integrations

MLS connectivity and automated property imports are the single highest-leverage capability gap between a standalone SaaS tool and a fully integrated investment platform. Unlocking live data reduces manual entry, increases platform stickiness, and positions the product as a source of truth rather than a calculation layer. A property data marketplace further opens a potential revenue stream through tiered API credits.

AI & Automation

The OpenAI integration established in V1 provides a proven inference pathway. V2 advances this by incorporating deal-scoring models, automated lead nurturing via email and SMS sequences, and contextual AI recommendations surfaced throughout the investor dashboard. Automation dramatically reduces the operational burden on the client team and increases the perceived intelligence of the platform.

Platform Expansion

Team accounts and investor collaboration unlock B2B2C monetization — allowing the client to sell seats to investment firms, syndicates, or coaching cohorts. A mobile app extends engagement outside the browser session, critical for deal flow management on the go. Investor Collaboration features (shared deal rooms, co-analysis views) differentiate the product in a competitive SaaS landscape.

CRM & Matching

The V1 CRM pipeline provides the data model and UI foundation. V2 evolves this into an intelligent matching engine — pairing buyer criteria with available properties, surfacing warm leads automatically, and enabling workflow automation across the sales cycle. Advanced CRM capabilities justify premium pricing tiers and dramatically improve user retention by embedding the platform into daily operations.

Phased Delivery Approach

V2 development is envisioned in three incremental waves, each building on the stability and user feedback from the prior release. Timelines are indicative and subject to prioritization discussions with the client.

1

V2.1 — Foundation & Data

~Months 1–4 post-launch

- MLS Integration
- Automated Property Imports
- Team Accounts (multi-tenancy)
- Advanced CRM enhancements

2

V2.2 — Intelligence & Automation

~Months 5–9 post-launch

- Advanced AI Deal Models
- Email & SMS Automation
- Buyer/Seller Matching Engine
- Property Data Marketplace (beta)

3

V2.3 — Scale & Collaboration

~Months 10–16 post-launch

- Mobile App (iOS & Android)
- Investor Collaboration (deal rooms)
- Property Data Marketplace (full)
- White-label / reseller tier

V1 Architectural Decisions Supporting V2

The following technical choices were made deliberately during V1 scoping to ensure V2 enhancements can be introduced without requiring a rewrite or major database migration.

1

Modular Service Layer

All business logic is encapsulated in isolated service modules (e.g., `dealService`, `propertyService`, `crmService`). New V2 capabilities can be introduced as additional service modules without touching existing logic, enabling clean feature branching and testability.

2

Prisma Schema Extensibility

The Neon/PostgreSQL schema is managed via Prisma with explicit relation definitions and nullable optional fields. V2 entities (e.g., MLS listings, team members, collaboration rooms) can be added as new models or schema extensions without altering existing tables, preserving backward compatibility.

3

Clerk Multi-Tenancy Support

Clerk's organization model is already configured in V1 but scoped to single-user flows. V2 team accounts require only enabling Clerk Organizations at the application level — the authentication scaffolding, JWT claims, and RBAC hooks are already present in the middleware layer.

4

Stripe Subscription Model Extensibility

V1 implements Stripe Billing with a clean product/price abstraction. Additional subscription tiers (team plans, usage-based credits for data marketplace, white-label reseller pricing) require only new Stripe Price objects and minor webhook handler extensions — no payment infrastructure rebuild.

5

6